

Classifying Tomorrow's Currents: A Probabilistic Neural Network Approach to Forecasting Electricity Prices

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Abstract—This study introduces a probabilistic neural network model for classifying next-day electricity prices into negative or positive categories. It addresses the challenge of price volatility exacerbated by renewable energy integration, presenting significant risks for market participants. Through a comparative analysis with the traditional support vector machine with radial basis function kernel using the German Day-Ahead market data of 2020, the probabilistic neural network demonstrates superior accuracy and yielded promising electricity price forecasting results. This approach equips market participants with valuable insights for strategic planning and risk mitigation.

Index Terms—Electricity price classification, probabilistic neural network, support vector machine, radial basis function, negative class, positive class

I. INTRODUCTION

The price of electricity exhibits volatility that may be attributed to various factors. A fundamental factor contributing to this volatility is the absence of cost-effective storage solutions [1]. Moreover, establishing a unified and equitable European market for electricity proved to be a challenging task, considering different geographical aspects, electricity flow directions and electricity supply patterns [2]. The absence of completely unified and equitable European markets also contributes to volatile electricity price behaviour.

Electricity prices are influenced by production from renewable sources, a phenomenon extensively explored in relevant literature. An in-depth review of studies analyzing the relationship between renewable energy production and electricity prices can be found in [3]. This summarises the performed research into three main groups: simulation-based studies, empirical studies, and studies

offering limited information on price effects. In [4], emphasis is placed on the significant role of energy production timing in determining electricity prices. This study also explains the correlation between the market value of renewables and other factors affecting prices. Positive correlations lead to increased market value, while negative correlations result in decreased value. Additionally, the study demonstrates that as renewables become more prevalent in the energy mix, their market value diminishes, aligning with the decline in energy prices according to the merit-order principle.

Furthermore, an empirical investigation into the interplay between renewable energy generation and wholesale prices was presented in [5]. The results of that study indicated that if the renewable energy share was averaged at 30% or 50% during the 2014-2021 period, prices would have experienced reductions of 8.8% and 20%, respectively. However, this transition would also induce significant volatility into the prices. Similar observations were reported in [6], in which the authors demonstrated that substantial wind energy production contributes to price reductions but amplifies market volatility.

Electricity prices are also shaped by various factors, including weather patterns, fuel prices, demand variability, and unexpected disruptions. Events such as the covid pandemic and the ongoing conflict in Ukraine have had a significant impact on energy prices [7], [8]. In the former case, prices plummeted to historic lows, while in the latter, they surged to unprecedented heights.

Consequently, participants in the electricity market find themselves vulnerable to significant risks that can impact their profitability. Hence, the utility of a forecasting tool for electricity prices is invaluable, as it enables these stakeholders to proactively manage and mitigate risks by furnishing them with vital information essential for formulating optimal bidding strategies,

as expounded in [9].

This paper introduces a methodology aimed at classifying electricity prices, with an emphasis on distinguishing between negative and positive price levels. More specifically, a probabilistic neural network (PNN) is developed to forecast the category of electricity prices for the following day. This information aids market participants in devising strategic decisions tailored to anticipated market conditions. This information helps market participants make strategic decisions tailored to anticipated market conditions. For instance, in the Day-Ahead market, where negative prices often rise with increased contributions from RES, a seller who wants to offer a quantity of energy should consider the forecast provided by the proposed methodology. If the proposed methodology forecasts negative prices for the next day, the seller should place a bid at lower prices to increase the likelihood of winning the auction, as the Day-Ahead market is governed by the principle that the selling price must be below the equilibrium price to secure a sale. Conversely, if the forecast indicates positive prices, the seller can confidently place higher bids.

On the demand side, if a buyer intends to purchase a quantity of energy and the methodology forecasts negative prices for the following day, they should prepare bids at relatively low prices. However, if positive prices are forecasted, the buyer should submit higher bids to avoid the risk of rejection, since winning the auction requires that the buying price exceeds the equilibrium price.

The performance of the PNN in classification was assessed and compared to that of a support vector machine (SVM) utilizing a radial basis function (RBF) kernel. The structure of the paper is as follows: Section II provides a review of existing probabilistic and classification methods found in the literature. Section III outlines the proposed PNN model, detailing its training and testing processes. Section IV discusses the implementation of the SVM with an RBF kernel. The results of the analysis are presented in Section V and in Section VI concludes the paper by providing a summary of the main findings, examining their significance, and outlining potential applications of the research.

II. LITERATURE SURVEY

The majority of existing research focuses on forecasting typical electricity prices [10]–[12]. Recent years have seen a growing emphasis on forecasting price outliers [10], particularly extreme price spikes influenced by the expanding integration of renewable energy sources into the electricity supply system. A probabilistic interval forecasting approach, utilizing extreme learning machine (ELM) was introduced in [11]. This method was evaluated across four distinct trading intervals using historical price and demand data from the Australian electricity market. The prediction interval coverage probability (PICP) for each test set was 94.13%, 93.36%, 95.70%, and 96.43%, respectively. In [12], a probabilistic price forecasting model based on the Nadaraya-Watson estimator and forecast probabilities from historical market price residuals was proposed. The model was tested on hourly imbalance settlement and Day-Ahead market prices from the Polish market. For the imbalance settlement, it yielded an average mean absolute percentage error (MAPE) of 20.35%, while for the Day-Ahead prices, the MAPE was 9.38%.

In [13], the authors presented a probabilistic forecasting method for electricity market prices using a Bayesian deep learning model, which includes a heteroscedasticity model for data

pre-processing. Their approach was tested over an entire year to evaluate monthly forecast performance, with parameters estimated from probability distributions within the model's weights. The data, comprising prices, load, and production, were sourced from the Belgian market. The methodology achieved a root mean square error (RMSE) of 11.58 (€/MWh). In another study [14], a combination of discrete cosine transforms (DCT) and cascade forward neural network (CFNN) for price classification was proposed. The training and testing data are locational marginal prices (LMPs) from the Spanish and New York ISO markets. The average mean percentage classification error (MPCE) for the Spanish market was 3.2%, while for New York, it was 4.1%.

In [15], the authors proposed a methodology for probabilistic price forecasting using variational heteroscedastic Gaussian process. They trained the model using historical price and load forecast data of 2013 from the Australian market, as well as market data from the PJM market in 2012 and the New England ISO market in 2011. The MAPE for each market was 4.44%, 8.06%, and 5.74%, respectively. Also, a new approach for forecasting distribution locational marginal prices using a spatio-temporal framework that integrates a deep bidirectional recurrent neural network was proposed in [16]. This method captures complex time dynamics and nodal price differentiation in large distribution systems. Their approach achieved a RMSE of 2.52 (c€/kWh). The authors in [17] explored probabilistic electricity price forecasting using nonlinear autoregressive with exogenous variables (NARX) networks, comparing the efficacy of combining point and probabilistic forecasts. The methodology employs datasets from the GEFCom2014 and Nord Pool power markets. The study compared various forecasting models, highlighting the superior performance of probabilistic forecasts derived directly from point predictions. For GEFCom2014, the approach yielded a weekly-weighted MAE (WMAE) of 14.72 (€/MWh), while for Nord Pool, it was 9.29 (€/MWh). A comprehensive study on forecasting electricity prices by incorporating probabilistic models such as autoregressive with exogenous variables (ARX) models with bivariate jumps and generalized autoregressive conditional heteroscedasticity (GARCH) a framework that considers the intricate dependency patterns between peak and off-peak price series was introduced in [18]. These models were applied to the Germany-Austria market, with the ARX model using ordinary least squares (OLS) yielding the best result, achieving a mean square error (MSE) of 1 (€/MWh) for both price series.

III. CLASSIFICATION MODEL: OVERVIEW

A. Probabilistic Neural Network (PNN) Model

The PNN model is not as commonly employed in applications as some other neural networks. Typically, it is employed in conjunction with other models, as exemplified in [19] and [20]. Additionally, PNN is frequently used as a classifier, providing valuable insights into the probability of spike occurrences, as highlighted in [21]. The rationale behind selecting the PNN model for our study lies in its ability to apply Bayes' rule to complicated data sets, enabling the computation of probability density functions (PDFs) that represent distinct classes.

The PNN architecture consists of four layers and it is depicted in Fig. 1, where its architecture is adapted for a two class classification problem. The first layer functions as the input layer, receiving the data to be classified. In the second layer, referred

to as the pattern layer, each neuron corresponds to a training example associated with a specific class. Here, Euclidean distances between the input data and the pattern layer are calculated, resulting in Gaussian mixtures that form the layer's output. The third layer, known as the summation layer, aggregates the PDF for each class. Finally, the fourth layer called the output layer, identifies the class with the highest probability, as described in [22].

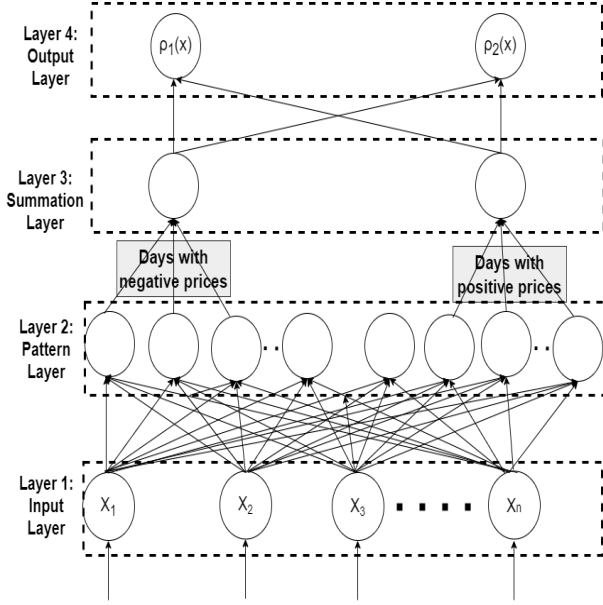


Fig. 1. PNN architecture for classifying electricity prices into negative and positive classes.

The PNN is trained using the maximum likelihood approach outlined in [23]. Consider an input data vector X of dimension Z , with M represents the number of distinct classes in the training dataset. The class-specific probability density function (pdf), $g_j(X)$ for class j , is derived using the mixture of probability density functions given by (1):

$$g_j(X) = \sum_{i=1}^{G_j} \pi_{ij} p_{ij}(X), \quad j = 1, 2, \dots, M \quad (1)$$

where G_j denotes the total count of Gaussian mixtures for class j , π_{ij} denotes the proportion of the i^{th} component within class j and p_{ij} refers to the multivariate probability density function given by (2):

$$p_{ij}(X) = (2\pi)^{-Z/2} |\Sigma|^{-1/2} e^{-1/2(X-\mu_{ij})^T \Sigma^{-1}(X-\mu_{ij})} \quad (2)$$

where Σ denotes the covariance matrix associated with the multivariate pdf, $p_{ij}(X)$, and μ_{ij} the mean. In the PNN fourth layer (see Fig. 1), the decision risk, denoted as $\rho_j(X)$, is computed to ascertain the minimal risk category for classifying the input vector X . This classification is done by calculating the decision risk using (3):

$$\rho_j(X) = \sum_{l=1}^M c_{jl} \alpha_l g_l(X) \quad (3)$$

where c_{jl} represents the cost of misclassifying the input vector X into class j when its correct classification is l , α_l indicates the prior probability of each class, reflecting their respective proportions within the dataset [23]. Each testing vector is classified in the class j for which the smallest risk ρ_j is computed.

B. Description of the training and testing methodology

This study utilizes data from the German Day-Ahead electricity market, specifically from 2020 [24], due to its open access nature and the occurrence of many negative prices triggered by Covid-19 lockdowns, providing a unique dataset for model training. If any day or even a single instance method recorded a negative price, it was categorized under the negative class. Conversely, days without any negative prices were classified as positive class. For model training and testing, a one-year period up to two days before a target day D was used for training, (i.e., $D - 365, D - 364, \dots, D - 2$ and the day immediately preceding D for testing, that is, $D - 1$).

As shown in (3), establishes that the decision risk, $\rho_j(X)$, calculation requires the misclassification cost, c_{jl} . In this study, the misclassification cost was assessed based on the operation of the Day-Ahead market, which employs two-sided auctions involving both supply and demand. Consequently, the PNN model was evaluated for accuracy on both fronts. On the supply side, if the model incorrectly assigns a negative class when it should be positive, the misclassification cost is assigned as the average price calculated from the training data in the negative class. Conversely, if the model mistakenly identifies a case as positive when it should be negative, the cost is based on the average price of the positive class training samples.

For demand scenarios, the misclassification costs are applied inversely compared to the supply side. If the model inaccurately predicts a negative class instead of the correct positive one, the cost is calculated using the average price of the positive class training samples. Similarly, if the model wrongfully predicts a positive class when the correct classification is negative, the misclassification cost aligns with the average price of the negative class training samples.

IV. EXPLORING SUPPORT VECTOR MACHINE (SVM) UTILIZING A RADIAL BASIS FUNCTION (RBF) KERNELS

The outcomes of PNN were compared to classification results obtained using traditional SVM. The SVM used in this study employed RBFs as kernels. While PNN utilizes data-induced probability density functions to define class boundaries, SVM typically employs a hyperplane to separate different classes. In situations where the data are not linearly separable, SVM utilizes kernel functions, such as the RBF, to effectively handle the non-linear relationships [25].

Consider a training dataset $(t_p, d_i)_p^H$ comprising H data points, where each t_p represents the p -th input vector, d_p represents the associated class label for the p -th observation. In scenarios where data exhibit non-linear relationships, the linear decision boundary of SVM is insufficient. The SVM utilizes the RBF kernel to map the original feature space into a higher-dimensional space, enabling the separation of classes by a hyperplane [25]. The decision function for an SVM using the RBF kernel can be represented as:

$$D(x) = \sum_{i=1}^H \alpha_p y_p K(t_p, t) + q \quad (4)$$

where $K(t_p, t) = e^{-\gamma \|t_p - t\|^2}$ is the RBF, γ is a parameter that determines the spread of the kernel, α_p is the coefficient of the support vector t_p and q is the bias. The coefficients α_p play a key role in determining the decision surface.

In addition to the RBF kernel, SVM employs slack variables ξ_i in the optimization problem as shown in (5), allowing some misclassifications or margin violations [26], [27]. This is particularly useful where perfect separation is not feasible, thus providing the necessary flexibility.

$$\min_{r, q, s} \frac{1}{2} \|r\|^2 + V \sum_{p=1}^H s_p \quad (5)$$

subject to

$$y_p(r^T \phi(x_p) + q) \geq 1 - s_p, \quad s_p \geq 0$$

where V is a penalty parameter that manages the balance between the error term and the regularization term, while s_i represents the distance from the margin [27] and $\phi(t_p)$ denotes the high-dimensional mapping of x_i induced by the RBF kernel [26].

V. RESULTS

In the German electricity market during 2020, there were 51 days with at least one occurrence of negative price and the remaining 315 days recorded only positive price levels. The performance of two classification models was assessed on their ability to accurately categorize each day based on varying training datasets. Their effectiveness was measured using an accuracy criterion, calculated as the proportion of correctly classified days to the total number of days in the respective class. Table I presents the PNN model's classification outcomes for the supply side. Meanwhile, Table II details the PNN model's results for the demand side, and Table III displays the classification performance of the SVM model.

TABLE I
PNN CLASSIFICATION RESULTS FOR SUPPLY SIDE

Training Dataset	Negative Class	Positive Class
Conventional Prod. For.	20 days (39.21%)	280 days (89.20%)
Forecasted Residual Load	15 days (29.41%)	281 days (89.21%)
Forecasted and Conventional Prod. for.	16 days (31.37%)	284 days (90.16%)
Production For. and RES For.	17 days (33.33%)	284 days (90.16%)

In the supply side scenario, the PNN model, trained with conventional production forecast data, identified the highest number of negative days—20 out of 51 days (39.21%). However, its ability to classify days into the positive class was somewhat inferior in comparison to other datasets. The most accurate results for the positive class, at 90.16%, were achieved using a combination of forecasted residual load data with both conventional production forecasts and renewable production forecast data.

Conversely, in the demand side scenario, the PNN model's performance for the negative class was not as strong as in

TABLE II
PNN CLASSIFICATION RESULTS FOR DEMAND SIDE

Training Dataset	Negative Class	Positive Class
Conventional Prod. For.	13 days (25.49%)	286 days (90.79%)
Forecasted Residual Load	12 days (23.53%)	305 days (96.82%)
Forecasted and Conventional Prod. for.	13 days (25.49%)	300 days (95.24%)
Production For. and RES For.	11 days (22.45%)	298 days (94.60%)

TABLE III
SVM CLASSIFICATION RESULTS

Training Dataset	Negative Class	Positive Class
Conventional Prod. For.	6 days (11.76%)	310 days (94.41%)
Forecasted Residual Load	17 days (33.33%)	300 days (95.24%)
Forecasted and Conventional Prod. for.	12 days (23.53%)	296 days (93.97%)
Production For. and RES For.	17 days (33.33%)	302 days (95.87%)

the supply side scenario, with the best accuracy being 25.49% using either conventional production forecast data alone or in combination with forecasted residual load data. Nevertheless, for the positive class, the model performed better in the demand side scenario; the use of forecasted residual load data alone led to an accuracy of 96.82%. Other combinations of data successfully identified nearly 300 of the 315 days with positive values, except when relying solely on conventional production forecast data, which identified 286 days.

Regarding the SVM model, it demonstrated comparable performance to the PNN in the supply side scenario for the negative class, especially with forecasted residual load data and the combination of production forecast with renewable production forecast data. While the PNN showed higher accuracy for the negative class with conventional production forecast data (supply side scenario), the SVM surpassed the PNN in positive class classification using the same data, achieving 94.41% accuracy. However, it also presented the lowest accuracy for the negative class at 11.76%.

VI. CONCLUSIONS

Electricity prices exhibit high volatility due to numerous influencing factors and the challenge of economical storage. The introduction of renewable energy has further increased market volatility, leading to significant financial risks for market participants. This paper explores the classification of electricity prices into positive and negative classes using the PNN model, providing forecasts that are crucial for market participants in devising optimal bidding strategies.

A key innovation of this study lies in the novel approach to determine misclassification costs, guided by the operational principles of the Day-Ahead market. This methodology introduces distinct misclassification costs for both supply and demand sides, tailored to market dynamics.

Additionally, this work includes a comparative analysis between the PNN and the conventional SVM model utilizing a

RBF kernel. While both models demonstrated comparable performance, the PNN exhibited marginally superior adaptability to negative price classifications, underscoring its potential resilience against the intricate patterns of electricity price volatility. Finally, this study paves the way for the development of a hybrid predictive framework (which will be part of future work), leveraging the strengths of PNN model to enhance forecasting accuracy in the volatile electricity market.

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